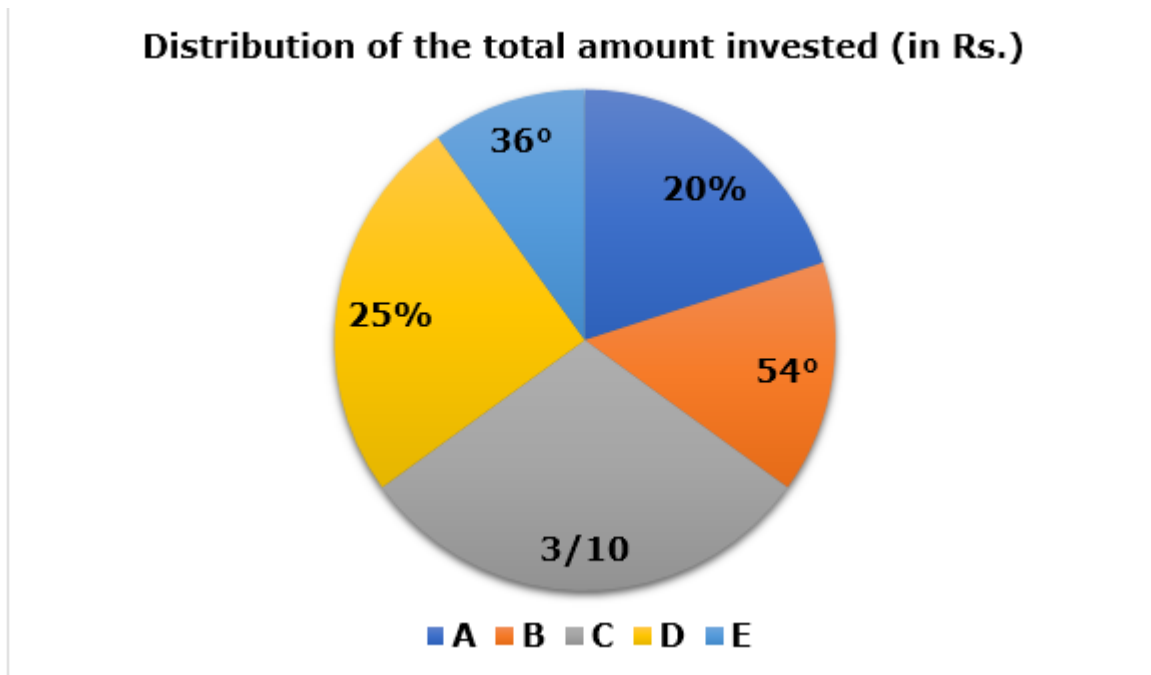


1. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the distribution of the total amount invested (in Rs.) by a person in five schemes A, B, C, D and E either in percentage, degree or fraction.



Notes:

- i). The total amount received from any scheme = the amount invested in the scheme + the interest received from the scheme
- ii). Only schemes B and C offer simple interest, while the rest of the schemes offer compound interest. (*compounded annually*)
- iii). The rate of interest (per annum) offered in schemes A, B, C, D and E is 20%, 12%, 18%, 25%, and 10% respectively.
- iv). The interest received from scheme A after 2 years is Rs. 8040 more than the interest received from scheme E after 2 years.
- v). The ratio of the time duration (in years) for which the amount has been invested in scheme B, C and D is 4:3:1.
- vi). The total amount received from schemes B and C is Rs. 82080.

Out of the total amount received from scheme A, Rs. [X] has been invested in another scheme at the rate of 15% per annum compounded annually for 2 years. If the same amount was invested at simple interest at same rate of interest for same time duration, then the simple interest received will be Rs. 450 less than the compound interest received. Find what percent (approx.) of the total amount received from scheme A is invested in another scheme?

- a. 75%
- b. 46%

- c. 92%
- d. 58%
- e. None of these

2. Questions

Out of the total amount received from Scheme D, a portion was invested in a bank at 5% per annum, compounded annually for 2 years, while the remaining amount was invested in a mutual fund scheme at a rate of 10% per annum simple interest for 4 years. The total interest received from both the bank and the mutual fund scheme was Rs. 8,306.25. Find the difference between the amounts invested in the bank and the mutual fund scheme

- a. Rs. 6000
- b. Rs. 7500
- c. Rs. 5000
- d. Cannot be determined
- e. None of these

3. Questions

If _____th of the amount invested in scheme B is further invested in a bank at the rate of _____% per annum compounded annually, then the compound interest received after 2 years will be Rs. _____.

Which of the following does satisfy the filler(s) in the same order?

I: $\frac{1}{4}$, 20 and 1980

II: $\frac{1}{9}$, 30 and 1440

III: $\frac{2}{5}$, 10 and 1512

- a. Only I follow
- b. Both II and III follow
- c. Both I and III follow
- d. All I, II and III follow
- e. None follow

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4. Questions

If Rs. [2500R] is invested in another scheme F at simple interest for 2 years, while the rate of interest for first and second year is [R]% per annum and [R+3]% per annum respectively, then the total interest received from scheme F after 2 years will be Rs. 8100. Find the ratio of the total amount invested in scheme E and F. (R is a positive integer)

- a. 2:5
- b. 5:2
- c. 3:1
- d. 4:3
- e. None of these

5. Questions

If the amount invested in scheme C is invested for [Y] years at same rate of interest (p.a.), then the simple interest received will be Rs. 12960. Find the compound interest received, if the same amount is invested at 20% per annum compounded annually for [Y] years.

- a. Rs. 12680
- b. Rs. 18260
- c. Rs. 14320
- d. Rs. 15840
- e. None of these

6. Questions

There are two series given below. Both series has one missing term as [P] and [Q]. Both series follow different pattern.

Series I: 6, 18, 9, 27, __[P]__, 40.50

Series II: 8, 5, 7, __[Q]__, 31, 82.50

Determine correct relation between [P] and [Q].

- a. $P > Q$
- b. $P < Q$
- c. $P \geq Q$
- d. $P \leq Q$
- e. $P = Q$

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7. Questions

There are two series given below. The series I has one wrong and series II has one missing term as [K]. The series II starts with the correct term that will replace the wrong term of series I. Both series follow same pattern.

Series I: 13, 17, 26, 50, 100, 221

Series II: ____, 55, 64, 89, __[K]__, 259

Which of the following is/are definitely true about [K]?

I: Perfect square.

II: Odd natural number.

III: Exactly divisible by 3.

- a. Both II and III
- b. Only III
- c. Only I
- d. All I, II and III
- e. None is true

8. Questions

There is a series given below, which has one missing term as [L].

Series: 40, 49, 31, 58, __[L]__, 67

If there is another series, which follows the same pattern starts with [L+1], then find 4th term of another series.

- a. 36
- b. 50
- c. 41
- d. 29
- e. None of these

9. Questions

There are three series given below. Each series has one wrong term. Each series follows same pattern.

Series I: 120, 147, 83, 208, -8, 327

Series II: 200, 227, 160, 288, 72, 415

Series III: 150, 177, 113, 236, 22, 365

The [A], [B], and [C] represents the correct terms that will replace the wrong terms of series I, series II, and series III respectively.

Which of the following equation(s) is/are correct?

I: $[(A+C)-B] = 410$

II: $[\sqrt{(B+6)}] = 13$

III: $[B-C] = 75$

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- a. Only I follow
- b. Both I and II follow
- c. Both II and III follow
- d. All I, II and III follow
- e. None follow

10. Questions

There are three series given below. Each series has one missing term and each series follows same pattern.

Series I: 8, 15, 44, __[X]__, 874, 5243

Series II: 5, 9, 26, 103, __[Y]__, 3083

Series III: 12, 23, __[Z]__, 271, 1354, 8123

The product of the digits of which of the following is highest among [X], [Y] and [Z]?

- a. X
- b. Y
- c. Z
- d. All X, Y and Z
- e. None of these

11. Questions

Read the following information carefully and answer the questions based on it.

Certain number of candidates who appeared in an interview in three different years 2017, 2018 and 2019.

2017: The total number of candidates who appeared in the interview is 800, out of which $\frac{1}{4}$ th qualified, while the rest did not qualify in the interview. The ratio of the number of males who qualified to the number of females who did not qualify in the interview is 3:4. The number of females who qualified is 12.50% of the number of males who did not qualify in the interview.

2018: 11.11% of the total number of candidates who appeared for the interview are qualified. The total number of candidates who did not qualify is 40 more than the previous year. The number of males who qualified is $\frac{1}{7}$ th of the number of males who did not qualify in the interview. The number of females who did not qualify is 11 times of the number of females who qualified in the interview.

2019: The total number of candidates who appeared in the interview is 220 less than the previous year. Out of the total number of candidates who appeared, 10% qualified in the interview. The ratio of the number of males who qualified and the number of females who did not qualify is 3:20. The number of females who qualified is 8% of the number of males who did not qualify.

Note:

- i). The total number of candidates who appeared in the interview = the number of candidates who qualified + the number of candidates who did not qualify
- ii). The total number of candidates who qualified = the number of males who qualified + the number of females who qualified
- iii). The total number of candidates who did not qualify = the number of males who did not qualify + the number of females who did not qualify

The total number of males who did not appear in the interview in 2017 is $[11X]$, which is 20 more than the total number of females who did not appear in the interview in the same year. If 20% of the total number of candidates (males + females) who applied in 2017, did not appear in the interview, then find the value of $[\sqrt{(X+6)}]$.

- a. 5
- b. 4
- c. 6
- d. 3
- e. None of these

12. Questions

In 2018, if the ratio of the number of qualified males who are graduates and the number of qualified females who are graduates is 15:4, while the number of qualified males who are post-graduates is 7 more than the number of qualified females who are post-graduates, then find the total number of qualified candidates (males + females) who are graduates in 2018.

- a. 57
- b. 23
- c. 42
- d. 29
- e. None of these

13. Questions

In 2020, if the ratio of the total number of candidates (males + females) who qualified and those who did not qualify is 3:1, then find the total number of candidates who appeared in the interview in 2020.

Which of the following statement(s) is/are sufficient to answer the question?

Statement I: The number of males who qualified in 2020 is 104 more than the number of males who did not qualify in the same year.

Statement II: The number of females who did not qualify in 2020 is 24, which is $\frac{1}{5}$ th of the number of females who qualified in the interview in the same year.

- a. Only statement I alone is sufficient
- b. Only statement II alone is sufficient
- c. Either statement I or II alone is sufficient
- d. Both statements I and II together are necessary
- e. Neither statement I nor II is sufficient

14. Questions

In 2019, out of the total number of candidates (males + females) who qualified, some candidates joined, while the rest did not join. If the number of qualified males who did not join is $\frac{3}{5}$ th of the number of qualified females who did not join, while the number of qualified males who joined is 12 more than the number of qualified females who joined, then find the ratio of the number of males candidates who joined and those males who did not join.

- a. 7:3
- b. 3:2
- c. 9:1
- d. Cannot be determined
- e. None of these

15. Questions

In 2018, if the total number of candidates who applied for the interview is $[20Y^2+20]$, out of which $[40Y]$ candidates did not appear, then the value of $[Y]$ will be positive integer, which is $___[?]___.$

What should come in place of question mark?

- Prime number
- Odd number
- Composite number
- Both (a) and (b)
- None of these

16. Questions

The following question has two quantities as Quantity I and Quantity II. You have to determine the relationship between them and give an answer accordingly.

Quantity I: Find the value of $\lceil \sqrt{X} \rceil$. The marked price of article A is Rs. 28000, article A is sold for a profit of 25% after allowing a discount of Rs. $[80X]$. The cost price of article B is Rs. $[120X]$, which is marked-up by 50% above its cost price and sold after allowing a discount of Rs. 2000. The selling price of article B is equals to the cost price of article A.

Quantity II: Find the value of $[N/10]$. The total number of articles purchased by a person is $[4N]$, out of which $3/4^{\text{th}}$ articles have been sold, while the rest remained unsold. The cost price of each article is Rs. 50, which is sold at a profit of 60%. The overall profit earned by the person from selling the articles is Rs. 2000.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

17. Questions

Quantity I: Find the value of $[20M]$. A and B started a business with an investment of Rs. 500 and Rs. 300. After $[M]$ months from start of the business, A increased his initial investment by 20%, while B doubles his initial investment. At the end of the year, A and B shared profit in the ratio of 4:3.

Quantity II: Find the value of $[N-300]$. P and Q started a business, with an investment of Rs. $[2N]$ and Rs. $[2N-200]$. After 9 months, A reduced his initial investment by 50%, while after another 1 month, B reduced his initial investment by Rs. 300. At the end of the year, P and Q shared profit in the ratio of 7:6.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

18. Questions

Quantity I: Find the age of the husband after 4 years from now, if the ratio of the present ages of husband and wife is 16:15. The couple got married $[5Y]$ years ago, while the present age of their son is $[Y]$ years. The sum of the ages of husband and wife at the time of their marriage was 52 years and the difference between the present ages of mother and son is 29 years.

Quantity II: Find the present age of C, if the present age of A is $[7Z]$ years and after 5 years from now, the age of B will equal to the present age of A. The present age of C is $[8Z]$ years, while the average of the present ages of A, B and C is 35 years.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

19. Questions

Quantity I: Find the length of the platform, if the time taken by train P, which length is 320 meters, to cross the platform is equal to the time taken by train Q, which length is $\frac{3}{4}$ of train P, to cross the same platform. The speed of train P is 44 m/sec, which is 22.22% more than train Q.

Quantity II: Find 60% of the length of the train, if the time taken by the train to cross the platform of length 150 meters is 5 seconds, while the same train can cross a man running with the speed of 10 m/sec in same direction in $(\frac{10}{3})$ seconds. (The speed of train is greater than the speed of the man)

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

20. Questions

Quantity I: Find the value of [P]. A vessel contains a mixture of [8P] ml, in which the quantity of milk and water is in the ratio of 5:3. If 50% of the initial mixture is taken out and replaced with 20 ml water, then the quantity of milk and water becomes equal.

Quantity II: Find the value of [Q]. A vessel contains [12Q] ml of pure milk. Now, if [3Q] ml milk is taken out and replaced with the same quantity of water, then the quantity of milk will become 60 ml more than water in the resultant mixture.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

Explanations:

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1. Questions

Let the total amount invested in all five schemes together be Rs. 100k.

So, the amount invested in scheme A = 20% of 100k = Rs. 20k

The rate of interest (p.a.) for scheme A = $[(20+20)+\{(20*20)/100\}]\% = (40+4)\% = 44\%$

So, the interest received from scheme A = 44% of 20k = Rs. 8.80k

The amount invested in scheme E = $\{(36/360)*100k\} = \text{Rs. } 10k$

The rate of interest (p.a.) for scheme E = $[(10+10)+\{(10*10)/100\}]\% = (20+1)\% = 21\%$

So, the interest received from scheme E = 21% of 10k = Rs. 2.10k

Now,

$$8.80k - 2.10k = 8040$$

$$6.70k = 8040$$

$$k = 1200$$

Therefore, the total amount invested in all five schemes together = 100k = $(100 \times 1200) = \text{Rs. } 120000$

Scheme A (C.I.)

The total amount invested = 20% of 120000 = Rs. 24000

The interest received = 8.80k = $(8.80 \times 1200) = \text{Rs. } 10560$

The total amount received = $(24000 + 10560) = \text{Rs. } 34560$

Scheme E (C.I.)

The total amount invested = 36% of 120000 = $\{(36/360) \times 120000\} = \text{Rs. } 12000$

The interest received = 2.10k = $(2.10 \times 1200) = \text{Rs. } 2520$

The total amount received = $(12000 + 2520) = \text{Rs. } 14520$

Now, let the time duration for which the amount invested in scheme B, C and D be 4t years, 3t years and t years respectively.

The amount invested in scheme B = 54% of 120000 = $\{(54/360) \times 120000\} = \text{Rs. } 18000$

So, the interest received from scheme B = $(12 \times 4t)\%$ of 18000 = 48t% of 18000 = Rs. 8640t

The total amount received from scheme B = Rs. $(18000 + 8640t)$

The amount invested in scheme C = 3/10 of 120000 = Rs. 36000

So, the interest received from scheme C = $(18 \times 3t)\%$ of 36000 = 54t% of 36000 = Rs. 19440t

The total amount received from scheme C = Rs. $(36000 + 19440t)$

Now,

$$(18000 + 8640t) + (36000 + 19440t) = 82080$$

$$54000 + 28080t = 82080$$

$$28080t = 28080$$

$$t = 1$$

Scheme B (S.I.)

The amount invested = Rs. 18000

The interest received = Rs. 8640t = $(8640 \times 1) = \text{Rs. } 8640$

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The total amount received = $(18000+8640t) = (18000+8640) = \text{Rs. } 26640$

Scheme C (S.I.)

The amount invested = Rs. 36000

The interest received from = $19440t = (19440*1) = \text{Rs. } 19440$

The total amount received = $(36000+19440t) = (36000+19440) = \text{Rs. } 55440$

Scheme D (C.I.)

The amount invested = 25% of 120000 = Rs. 30000

The time duration = $t = 1$ year

The rate of interest (p.a.) = 25%

The interest received from = $(25*1)\%$ of 30000 = 25% of 30000 = Rs. 7500

The total amount received = $(30000+7500) = \text{Rs. } 37500$

Scheme	The total amount invested (in Rs.)	The rate of interest (per annum)	The time duration (in years)	The interest received (in Rs.)	The total amount received (in Rs.)
A (C.I.)	24000	20%	2	10560	34560
B (S.I.)	18000	12%	4	8640	26640
C (S.I.)	36000	18%	3	19440	55440
D (C.I.)	30000	25%	1	7500	37500
E (C.I.)	12000	10%	2	2520	14520

Answer: D

Now, according to the question,

The total amount invested in another scheme = Rs. X

The time duration = 2 years

The rate of interest (p.a.), if compounded annually = $[(15+15)+\{(15*15)/100\}]\% = (30+2.25)\% = 32.25\%$

The rate of interest (p.a.) at simple interest = $(15*2)\% = 30\%$ @QuizByMickey

So, the compound interest received = 32.25% of X = Rs. 0.3225X

And, the simple interest received = 30% of X = Rs. 0.30X

Now,

$$0.3225X - 0.30X = 450$$

$$0.0225X = 450$$

$$X = 20000$$

Also, the total amount received from scheme A = Rs. 34560

And, the amount invested in another scheme = X = Rs. 20000

Therefore, the required percentage = $\{(20000/34560) \times 100\} \% = 57.87\% \approx 58\%$.

Hence, the correct answer is option D.

2. Questions

Let the total amount invested in all five schemes together be Rs. 100k.

So, the amount invested in scheme A = 20% of 100k = Rs. 20k

The rate of interest (p.a.) for scheme A = $[(20+20) + \{(20 \times 20)/100\}] \% = (40+4)\% = 44\%$

So, the interest received from scheme A = 44% of 20k = Rs. 8.80k

The amount invested in scheme E = $\{(36/360) \times 100k\} = \text{Rs. } 10k$

The rate of interest (p.a.) for scheme E = $[(10+10) + \{(10 \times 10)/100\}] \% = (20+1)\% = 21\%$

So, the interest received from scheme E = 21% of 10k = Rs. 2.10k

Now,

$$8.80k - 2.10k = 8040$$

$$6.70k = 8040$$

$$k = 1200$$

Therefore, the total amount invested in all five schemes together = 100k = $(100 \times 1200) = \text{Rs. } 120000$

Scheme A (C.I.)

The total amount invested = 20% of 120000 = Rs. 24000

The interest received = 8.80k = $(8.80 \times 1200) = \text{Rs. } 10560$

The total amount received = $(24000 + 10560) = \text{Rs. } 34560$

Scheme E (C.I.)

The total amount invested = 36% of 120000 = $\{(36/360) \times 120000\} = \text{Rs. } 12000$

The interest received = 2.10k = $(2.10 \times 1200) = \text{Rs. } 2520$

The total amount received = $(12000 + 2520) = \text{Rs. } 14520$

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Now, let the time duration for which the amount invested in scheme B, C and D be 4t years, 3t years and t years respectively.

The amount invested in scheme B = 54% of 120000 = $\{(54/360) \times 120000\} = \text{Rs. } 18000$

So, the interest received from scheme B = $(12 \times 4t)\%$ of 18000 = $48t\%$ of 18000 = Rs. 8640t

The total amount received from scheme B = Rs. $(18000 + 8640t)$

The amount invested in scheme C = $3/10$ of 120000 = Rs. 36000

So, the interest received from scheme C = $(18 \times 3t)\%$ of 36000 = $54t\%$ of 18000 = Rs. 19440t

The total amount received from scheme C = Rs. (36000+19440t)

Now,

$$(18000+8640t) + (36000+19440t) = 82080$$

$$54000 + 28080t = 82080$$

$$28080t = 28080$$

$$t = 1$$

Scheme B (S.I.)

The amount invested = Rs. 18000

The interest received = Rs. 8640t = (8640×1) = Rs. 8640

The total amount received = $(18000+8640t) = (18000+8640)$ = Rs. 26640

Scheme C (S.I.)

The amount invested = Rs. 36000

The interest received from = 19440t = (19440×1) = Rs. 19440

The total amount received = $(36000+19440t) = (36000+19440)$ = Rs. 55440

Scheme D (C.I.)

The amount invested = 25% of 120000 = Rs. 30000

The time duration = t = 1 year

The rate of interest (p.a.) = 25%

The interest received from = $(25 \times 1)\%$ of 30000 = 25% of 30000 = Rs. 7500

The total amount received = $(30000+7500)$ = Rs. 37500

Scheme	The total amount invested (in Rs.)	The rate of interest (per annum)	The time duration (in years)	The interest received (in Rs.)	The total amount received (in Rs.)
A (C.I.)	24000	20%	2	10560	34560
B (S.I.)	18000	12%	4	8640	26640
C (S.I.)	36000	18%	3	19440	55440
D (C.I.)	30000	25%	1	7500	37500
E (C.I.)	12000	10%	2	2520	14520

Answer: B

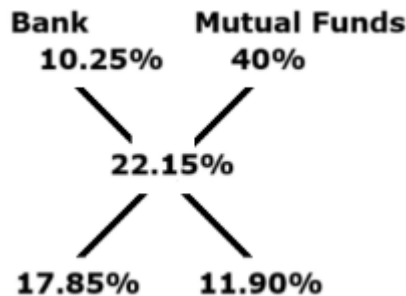
The total amount received from scheme D = Rs. 37500

The rate of interest in the bank (C.I.) = $[(5+5)+\{(5 \times 5)/100\}]\% = (10+0.25)\% = 10.25\%$ p.a. (compounded annually)

The rate of interest in the mutual funds scheme (S.I.) = $(10 \times 4)\% = 40\%$ p.a.

The overall rate of interest (bank + mutual funds scheme) = $\{(8306.25/37500) \times 100\}\% = 22.15\%$

By using allegation method, we get,



So, the ratio of the amount invested in the bank and mutual funds scheme = $17.85\%:11.90\% = 3:2$.

Therefore, the difference between the amount invested in the bank and mutual funds scheme = $1/5$ of $37500 = \text{Rs. } 7500$.

Hence, the correct answer is option B.

3. Questions

Let the total amount invested in all five schemes together be Rs. 100k.

So, the amount invested in scheme A = 20% of $100\text{k} = \text{Rs. } 20\text{k}$

The rate of interest (p.a.) for scheme A = $[(20+20)+\{(20 \times 20)/100\}]\% = (40+4)\% = 44\%$

So, the interest received from scheme A = 44% of $20\text{k} = \text{Rs. } 8.80\text{k}$

The amount invested in scheme E = $\{(36/360) \times 100\text{k}\} = \text{Rs. } 10\text{k}$

The rate of interest (p.a.) for scheme E = $[(10+10)+\{(10 \times 10)/100\}]\% = (20+1)\% = 21\%$

So, the interest received from scheme E = 21% of $10\text{k} = \text{Rs. } 2.10\text{k}$

Now,

$$8.80\text{k} - 2.10\text{k} = 8040$$

$$6.70\text{k} = 8040$$

$$k = 1200$$

Therefore, the total amount invested in all five schemes together = $100\text{k} = (100 \times 1200) = \text{Rs. } 120000$

Scheme A (C.I.)

The total amount invested = 20% of $120000 = \text{Rs. } 24000$

The interest received = $8.80\text{k} = (8.80 \times 1200) = \text{Rs. } 10560$

The total amount received = $(24000 + 10560) = \text{Rs. } 34560$

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Scheme E (C.I.)

The total amount invested = 36° of 120000 = $\{(36/360)*120000\}$ = Rs. 12000

The interest received = $2.10k = (2.10*1200) =$ Rs. 2520

The total amount received = $(12000+2520) =$ Rs. 14520

Now, let the time duration for which the amount invested in scheme B, C and D be $4t$ years, $3t$ years and t years respectively.

The amount invested in scheme B = 54° of 120000 = $\{(54/360)*120000\} =$ Rs. 18000

So, the interest received from scheme B = $(12*4t)\%$ of 18000 = $48t\%$ of 18000 = Rs. $8640t$

The total amount received from scheme B = Rs. $(18000+8640t)$

The amount invested in scheme C = $3/10$ of 120000 = Rs. 36000

So, the interest received from scheme C = $(18*3t)\%$ of 36000 = $54t\%$ of 18000 = Rs. $19440t$

The total amount received from scheme C = Rs. $(36000+19440t)$

Now,

$$(18000+8640t) + (36000+19440t) = 82080$$

$$54000 + 28080t = 82080$$

$$28080t = 28080$$

$$t = 1$$

Scheme B (S.I.)

The amount invested = Rs. 18000

The interest received = Rs. $8640t = (8640*1) =$ Rs. 8640

The total amount received = $(18000+8640t) = (18000+8640) =$ Rs. 26640

Scheme C (S.I.)

The amount invested = Rs. 36000

The interest received from = $19440t = (19440*1) =$ Rs. 19440

The total amount received = $(36000+19440t) = (36000+19440) =$ Rs. 55440

Scheme D (C.I.)

The amount invested = 25% of 120000 = Rs. 30000

The time duration = $t = 1$ year

The rate of interest (p.a.) = 25%

The interest received from = $(25*1)\%$ of 30000 = 25% of 30000 = Rs. 7500

The total amount received = $(30000+7500) =$ Rs. 37500

Scheme	The total amount invested (in Rs.)	The rate of interest (per annum)	The time duration (in years)	The interest received (in Rs.)	The total amount received (in Rs.)
A (C.I.)	24000	20%	2	10560	34560
B (S.I.)	18000	12%	4	8640	26640
C (S.I.)	36000	18%	3	19440	55440
D (C.I.)	30000	25%	1	7500	37500
E (C.I.)	12000	10%	2	2520	14520

Answer: C

The total amount invested in scheme B = Rs. 18000

Using I

The amount invested = $1/4$ of 18000 = Rs. 4500

The rate of interest = $[(20+20)+\{(20*20)/100\}]\% = (40+4)\% = 44\%$

So, the compound interest received after 2 years = 44% of 4500 = Rs. 1980.

Therefore, I does satisfy.

Using II

The amount invested = $1/9$ of 18000 = Rs. 2000

The rate of interest = $[(30+30)+\{(30*30)/100\}]\% = (60+9)\% = 69\%$

So, the compound interest received after 2 years = 69% of 2000 = Rs. 1380.

Therefore, II does not satisfy.

Using III

The amount invested = $2/5$ of 18000 = Rs. 7200

The rate of interest = $[(10+10)+\{(10*10)/100\}]\% = (20+1)\% = 21\%$

So, the compound interest received after 2 years = 21% of 7200 = Rs. 1512.

Therefore, III does satisfy.

So, both I and III follow.

Hence, the correct answer is option C.

4. Questions

Let the total amount invested in all five schemes together be Rs. 100k.

So, the amount invested in scheme A = 20% of 100k = Rs. 20k

The rate of interest (p.a.) for scheme A = $[(20+20)+\{(20*20)/100\}]\% = (40+4)\% = 44\%$

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So, the interest received from scheme A = 44% of 20k = Rs. 8.80k

The amount invested in scheme E = $\{(36/360)*100k\}$ = Rs. 10k

The rate of interest (p.a.) for scheme E = $[(10+10)+\{(10*10)/100\}] \% = (20+1)\% = 21\%$

So, the interest received from scheme E = 21% of 10k = Rs. 2.10k

Now,

$$8.80k - 2.10k = 8040$$

$$6.70k = 8040$$

$$k = 1200$$

Therefore, the total amount invested in all five schemes together = 100k = (100×1200) = Rs. 120000

Scheme A (C.I.)

The total amount invested = 20% of 120000 = Rs. 24000

The interest received = 8.80k = (8.80×1200) = Rs. 10560

The total amount received = $(24000 + 10560)$ = Rs. 34560

Scheme E (C.I.)

The total amount invested = 36% of 120000 = $\{(36/360)*120000\}$ = Rs. 12000

The interest received = 2.10k = (2.10×1200) = Rs. 2520

The total amount received = $(12000 + 2520)$ = Rs. 14520

Now, let the time duration for which the amount invested in scheme B, C and D be 4t years, 3t years and t years respectively.

The amount invested in scheme B = 54% of 120000 = $\{(54/360)*120000\}$ = Rs. 18000

So, the interest received from scheme B = $(12*4t)\%$ of 18000 = 48t% of 18000 = Rs. 8640t

The total amount received from scheme B = Rs. $(18000 + 8640t)$

The amount invested in scheme C = 3/10 of 120000 = Rs. 36000

So, the interest received from scheme C = $(18*3t)\%$ of 36000 = 54t% of 36000 = Rs. 19440t

The total amount received from scheme C = Rs. $(36000 + 19440t)$

Now,

$$(18000 + 8640t) + (36000 + 19440t) = 82080$$

$$54000 + 28080t = 82080$$

$$28080t = 28080$$

$$t = 1$$

Scheme B (S.I.)

The amount invested = Rs. 18000

The interest received = Rs. 8640 $t = (8640 \times 1) = \text{Rs. } 8640$

The total amount received = $(18000 + 8640) = (18000 + 8640) = \text{Rs. } 26640$

Scheme C (S.I.)

The amount invested = Rs. 36000

The interest received from = $19440t = (19440 \times 1) = \text{Rs. } 19440$

The total amount received = $(36000 + 19440) = (36000 + 19440) = \text{Rs. } 55440$

Scheme D (C.I.)

The amount invested = 25% of 120000 = Rs. 30000

The time duration = $t = 1$ year

The rate of interest (p.a.) = 25%

The interest received from = $(25 \times 1)\%$ of 30000 = 25% of 30000 = Rs. 7500

The total amount received = $(30000 + 7500) = \text{Rs. } 37500$

Scheme	The total amount invested (in Rs.)	The rate of interest (per annum)	The time duration (in years)	The interest received (in Rs.)	The total amount received (in Rs.)
A (C.I.)	24000	20%	2	10560	34560
B (S.I.)	18000	12%	4	8640	26640
C (S.I.)	36000	18%	3	19440	55440
D (C.I.)	30000	25%	1	7500	37500
E (C.I.)	12000	10%	2	2520	14520

Answer: A

Scheme F (S.I.)

The total amount invested = Rs. 2500R

So, the interest received after first year = $R\%$ of 2500R = Rs. $25R^2$

And, the interest received after second year = $(R+3)\%$ of 2500R = Rs. $(25R^2 + 75R)$

Therefore, the total interest received after 2 years = $\{25R^2 + (25R^2 + 75R)\} = \text{Rs. } (50R^2 + 75R)$

Now,

$$50R^2 + 75R = 8100$$

$$50R^2 + 75R - 8100 = 0$$

$$2R^2 + 3R - 324 = 0$$

$$2R^2 + 27R - 24R - 324 = 0$$

$$2R(R+13.50) - 24(R+13.50) = 0$$

$$(2R-24)(R+13.50) = 0$$

$$R = +12 \text{ or } -13.50$$

So, $R = 12$ (positive integer)

Now, the total amount invested in scheme F = $2500R = (2500 \times 12) = \text{Rs. } 30000$

And, the total amount invested in scheme E = Rs. 12000

Therefore, the required ratio = $12000:30000 = 2:5$.

Hence, the correct answer is option A.

5. Questions

Let the total amount invested in all five schemes together be Rs. 100k.

So, the amount invested in scheme A = 20% of 100k = Rs. 20k

The rate of interest (p.a.) for scheme A = $[(20+20) + \{(20 \times 20)/100\}] \% = (40+4)\% = 44\%$

So, the interest received from scheme A = 44% of 20k = Rs. 8.80k

The amount invested in scheme E = $\{(36/360) \times 100k\} = \text{Rs. } 10k$

The rate of interest (p.a.) for scheme E = $[(10+10) + \{(10 \times 10)/100\}] \% = (20+1)\% = 21\%$

So, the interest received from scheme E = 21% of 10k = Rs. 2.10k

Now,

$$8.80k - 2.10k = 8040$$

$$6.70k = 8040$$

$$k = 1200$$

Therefore, the total amount invested in all five schemes together = $100k = (100 \times 1200) = \text{Rs. } 120000$

Scheme A (C.I.)

The total amount invested = 20% of 120000 = Rs. 24000

The interest received = $8.80k = (8.80 \times 1200) = \text{Rs. } 10560$

The total amount received = $(24000 + 10560) = \text{Rs. } 34560$

Scheme E (C.I.)

The total amount invested = 36% of 120000 = $\{(36/360) \times 120000\} = \text{Rs. } 12000$

The interest received = $2.10k = (2.10 \times 1200) = \text{Rs. } 2520$

The total amount received = $(12000 + 2520) = \text{Rs. } 14520$

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Now, let the time duration for which the amount invested in scheme B, C and D be $4t$ years, $3t$ years and t years respectively.

The amount invested in scheme B = 54% of $120000 = \{(54/360)*120000\} = \text{Rs. } 18000$

So, the interest received from scheme B = $(12*4t)\%$ of $18000 = 48t\%$ of $18000 = \text{Rs. } 8640t$

The total amount received from scheme B = $\text{Rs. } (18000+8640t)$

The amount invested in scheme C = $3/10$ of $120000 = \text{Rs. } 36000$

So, the interest received from scheme C = $(18*3t)\%$ of $36000 = 54t\%$ of $18000 = \text{Rs. } 19440t$

The total amount received from scheme C = $\text{Rs. } (36000+19440t)$

Now,

$$(18000+8640t) + (36000+19440t) = 82080$$

$$54000 + 28080t = 82080$$

$$28080t = 28080$$

$$t = 1$$

Scheme B (S.I.)

The amount invested = $\text{Rs. } 18000$

The interest received = $\text{Rs. } 8640t = (8640*1) = \text{Rs. } 8640$

The total amount received = $(18000+8640t) = (18000+8640) = \text{Rs. } 26640$

Scheme C (S.I.)

The amount invested = $\text{Rs. } 36000$

The interest received from = $19440t = (19440*1) = \text{Rs. } 19440$

The total amount received = $(36000+19440t) = (36000+19440) = \text{Rs. } 55440$

Scheme D (C.I.)

The amount invested = 25% of $120000 = \text{Rs. } 30000$

The time duration = $t = 1$ year

The rate of interest (p.a.) = 25%

The interest received from = $(25*1)\%$ of $30000 = 25\%$ of $30000 = \text{Rs. } 7500$

The total amount received = $(30000+7500) = \text{Rs. } 37500$

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Scheme	The total amount invested (in Rs.)	The rate of interest (per annum)	The time duration (in years)	The interest received (in Rs.)	The total amount received (in Rs.)
A (C.I.)	24000	20%	2	10560	34560
B (S.I.)	18000	12%	4	8640	26640
C (S.I.)	36000	18%	3	19440	55440
D (C.I.)	30000	25%	1	7500	37500
E (C.I.)	12000	10%	2	2520	14520

Answer: D

Scheme C

The total amount invested = Rs. 36000

So, the simple interest received = $(18 \times Y)\%$ of 36000 = $18Y\%$ of 36000 = Rs. 6480Y

Now,

$$6480Y = 12960$$

$$Y = 2$$

So, the time duration = $Y = 2$ years

Now, the rate of interest (compounded annually) = $[(20+20)+\{(20 \times 20)/100\}]\% = (40+4)\% = 44\%$ p.a.

Therefore, the compound interest received = 44% of 36000 = Rs. 15840.

Hence, the correct answer is option D.

6. Questions

Answer: E

Series I: 6, 18, 9, 27, __[P]__, 40.5

$$1^{\text{st}} \text{ term} = 6$$

$$2^{\text{nd}} \text{ term} = (6 \times 3) = 18$$

$$3^{\text{rd}} \text{ term} = (18/2) = 9$$

$$4^{\text{th}} \text{ term} = (9 \times 3) = 27$$

$$5^{\text{th}} \text{ term} = (27/2) = 13.50 \text{ (P)}$$

$$6^{\text{th}} \text{ term} = (13.50 \times 3) = 40.50$$

So, the missing term = $P = 13.50$

Series II: 8, 5, 7, __[Q]__, 31, 82.50

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$$1^{\text{st}} \text{ term} = 8$$

$$2^{\text{nd}} \text{ term} = (8 \times 0.50 + 1) = (4 + 1) = 5$$

$$3^{\text{rd}} \text{ term} = (5 \times 1 + 2) = (5 + 2) = 7$$

$$3^{\text{rd}} \text{ term} = (7 \times 1.50 + 3) = (10.50 + 3) = 13.50 \text{ (Q)}$$

$$4^{\text{th}} \text{ term} = (13.50 \times 2 + 4) = (27 + 4) = 31$$

$$5^{\text{th}} \text{ term} = (31 \times 2.50 + 5) = (77.50 + 5) = 82.50$$

So, the missing term = Q = 13.50

Therefore, P = Q.

Hence, the correct answer is option E.

7. Questions

Answer: B

Series I: 13, 17, 26, 50, 100, 221

$$1^{\text{st}} \text{ term} = 13$$

$$2^{\text{nd}} \text{ term} = (13 + 2^2) = (13 + 4) = 17$$

$$3^{\text{rd}} \text{ term} = (17 + 3^2) = (17 + 9) = 26$$

$$4^{\text{th}} \text{ term} = (26 + 5^2) = (26 + 25) = 51 \text{ (50 is wrong)}$$

$$5^{\text{th}} \text{ term} = (51 + 7^2) = (51 + 49) = 100$$

$$6^{\text{th}} \text{ term} = (100 + 11^2) = (100 + 121) = 221$$

So, the wrong term = 50

And, the correct term = 51

Series II: ____, 55, 64, 89, __[K]__, 259

$$1^{\text{st}} \text{ term} = 51$$

$$2^{\text{nd}} \text{ term} = (51 + 2^2) = (51 + 4) = 55$$

$$3^{\text{rd}} \text{ term} = (55 + 3^2) = (55 + 9) = 64$$

$$4^{\text{th}} \text{ term} = (64 + 5^2) = (64 + 25) = 89$$

$$5^{\text{th}} \text{ term} = (89 + 7^2) = (89 + 49) = 138 \text{ (missing term)}$$

$$6^{\text{th}} \text{ term} = (138 + 11^2) = (138 + 121) = 259$$

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So, the missing term = $K = 138$

138 neither perfect square nor odd natural number. But, 138 is exactly divisible by 3.

$$(138/3) = 46$$

Therefore, only III is definitely true.

Hence, the correct answer is option B.

8. Questions

Answer: C

Series: 40, 49, 31, 58, __[L]__, 67

$$1^{\text{st}} \text{ term} = 40$$

$$2^{\text{nd}} \text{ term} = (40+9) = 49$$

$$3^{\text{rd}} \text{ term} = (49-18) = 31$$

$$4^{\text{th}} \text{ term} = (31+27) = 58$$

$$5^{\text{th}} \text{ term} = (58-36) = 22 \text{ (missing)}$$

$$6^{\text{th}} \text{ term} = (22+45) = 67$$

So, the missing term = $L = 22$

Another Series

$$1^{\text{st}} \text{ term} = (L+1) = (22+1) = 23$$

$$2^{\text{nd}} \text{ term} = (23+9) = 32$$

$$3^{\text{rd}} \text{ term} = (32-18) = 14$$

$$4^{\text{th}} \text{ term} = (14+27) = 41$$

$$5^{\text{th}} \text{ term} = (41-36) = 5$$

$$6^{\text{th}} \text{ term} = (5+45) = 50$$

Therefore, the 4th term of another series = 41.

Hence, the correct answer is option C.

9. Questions

Answer: B

Series I: 120, 147, 83, 208, -8, 327

$$1^{\text{st}} \text{ term} = 120$$

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$$2^{\text{nd}} \text{ term} = (120+3^3) = (120+27) = 147$$

$$3^{\text{rd}} \text{ term} = (147-4^3) = (147-64) = 83$$

$$4^{\text{th}} \text{ term} = (83+5^3) = (83+125) = 208$$

$$5^{\text{th}} \text{ term} = (208-6^3) = (208-216) = -8$$

$$6^{\text{th}} \text{ term} = (-8+7^3) = (-8+343) = 335 \text{ (327 is wrong)}$$

So, the wrong term = 327

And, the correct term = A = 335

Series II: 200, 227, 160, 288, 72, 415

$$1^{\text{st}} \text{ term} = 200$$

$$2^{\text{nd}} \text{ term} = (200+3^3) = (200+27) = 227$$

$$3^{\text{rd}} \text{ term} = (227-4^3) = (227-64) = 163 \text{ (160 is wrong)}$$

$$4^{\text{th}} \text{ term} = (163+5^3) = (163+125) = 288$$

$$5^{\text{th}} \text{ term} = (288-6^3) = (288-216) = 72$$

$$6^{\text{th}} \text{ term} = (72+7^3) = (72+343) = 415$$

So, the wrong term = 160

And, the correct term = B = 163

Series III: 150, 177, 113, 236, 22, 365

$$1^{\text{st}} \text{ term} = 150$$

$$2^{\text{nd}} \text{ term} = (150+3^3) = (150+27) = 177$$

$$3^{\text{rd}} \text{ term} = (177-4^3) = (177-64) = 113$$

$$4^{\text{th}} \text{ term} = (113+5^3) = (113+125) = 238 \text{ (236 is wrong)}$$

$$5^{\text{th}} \text{ term} = (238-6^3) = (238-216) = 22$$

$$6^{\text{th}} \text{ term} = (22+7^3) = (22+343) = 365$$

So, the wrong term = 236

And, the correct term = C = 238

$$\text{I: } \{(A+C)-B\} = \{(335+238)-163\} = (573-163) = 410$$

So, I is correct.

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$$\text{II: } \sqrt{(B+6)} = \sqrt{(163+6)} = \sqrt{169} = 13$$

So, II is correct.

$$\text{III: } (B-C) = (163-238) = -75$$

So, III is not correct.

Therefore, both I and II follow.

Hence, the correct answer is option B.

10. Questions

Answer: C

Series I: 8, 15, 44, __[X]__, 874, 5243

$$1^{\text{st}} \text{ term} = 8$$

$$2^{\text{nd}} \text{ term} = (8*2-1) = (16-1) = 15$$

$$3^{\text{rd}} \text{ term} = (15*3-1) = (45-1) = 44$$

$$4^{\text{th}} \text{ term} = (44*4-1) = (176-1) = 175 \text{ (missing)}$$

$$5^{\text{th}} \text{ term} = (175*5-1) = (875-1) = 874$$

$$6^{\text{th}} \text{ term} = (874*6-1) = (5244-1) = 5243$$

So, the missing term = X = 175

Series II: 5, 9, 26, 103, __[Y]__, 3083

$$1^{\text{st}} \text{ term} = 5$$

$$2^{\text{nd}} \text{ term} = (5*2-1) = (10-1) = 9$$

$$3^{\text{rd}} \text{ term} = (9*3-1) = (27-1) = 26$$

$$4^{\text{th}} \text{ term} = (26*4-1) = (104-1) = 103$$

$$5^{\text{th}} \text{ term} = (103*5-1) = (515-1) = 514 \text{ (missing)}$$

$$6^{\text{th}} \text{ term} = (514*6-1) = (3084-1) = 3083$$

So, the missing term = Y = 514

Series III: 12, 23, __[Z]__, 271, 1354, 8123

$$1^{\text{st}} \text{ term} = 12$$

$$2^{\text{nd}} \text{ term} = (12*2-1) = (24-1) = 23$$

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$$3^{\text{rd}} \text{ term} = (23 \times 3 - 1) = (69 - 1) = 68$$

$$4^{\text{th}} \text{ term} = (68 \times 4 - 1) = (272 - 1) = 271$$

$$5^{\text{th}} \text{ term} = (271 \times 5 - 1) = (1355 - 1) = 1354$$

$$6^{\text{th}} \text{ term} = (1354 \times 6 - 1) = (8124 - 1) = 8123$$

So, the missing term = Z = 68

Now, the product of the digits of,

$$X = 175 = (1 \times 7 \times 5) = 35$$

$$Y = 514 = (5 \times 1 \times 4) = 20$$

$$Z = 68 = (6 \times 8) = 48 \text{ (highest)}$$

Therefore, the product of the digits of Z is highest among all three.

Hence, the correct answer is option C.

11. Questions

2017

The total number of candidates who appeared = 800

The total number of candidates who qualified = $\frac{1}{4}$ of 800 = 200

The total number of candidates who did not qualify = $\frac{3}{4}$ of 800 = 600

Now, let the number of males who qualified and females who did not qualify be 3m and 4m respectively.

Also, the ratio of the number of females who qualified and males who did not qualify = 1:8. Let it be n and 8n respectively. (**Note: 12.50% = $\frac{1}{8}$**)

So,

$$3m + n = 200 \dots \dots \dots \text{(I)}$$

And,

$$4m + 8n = 600$$

$$m + 2n = 150 \dots \dots \dots \text{(II)}$$

By solving equations (I) and (II), we get,

$$3m + n = 200$$

$$3m + 6n = 450$$

$$5n = 250$$

$$n = 50$$

Now, put the value of n in equation (I), we get,

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$$3m + n = 200$$

$$3m + 50 = 200$$

$$3m = 150$$

$$m = 50$$

Therefore,

$$\text{The number males who qualified} = 3m = (3 \times 50) = 150$$

$$\text{The number of females who qualified} = n = 50$$

$$\text{The number of males who did not qualify} = 8n = (8 \times 50) = 400$$

$$\text{The number of females who did not qualify} = 4m = (4 \times 50) = 200$$

2018

$$\text{The total number of candidates who did not qualify} = (600 + 40) = 640$$

The ratio of the total number of candidates who qualified and those who did not qualify = 1:8 (*Note: 11.11% = 1/9*)

$$\text{The total number of candidates who qualified} = \{(640/8) \times 1\} = 80$$

$$\text{The total number of candidates who appeared} = (80 + 640) = 720$$

The ratio of the number of males who qualified and males who did not qualify = 1:7. Let it be a and $7a$ respectively.

Also, the ratio of the number of females who qualified and females who did not qualify = 1:11. Let it be b and $11b$ respectively.

Now,

$$a + b = 80 \dots \dots \dots \text{(III)}$$

$$7a + 11b = 640 \dots \dots \dots \text{(IV)}$$

By solving equation (I) and (IV), we get,

$$7a + 7b = 560$$

$$7a + 11b = 640$$

$$4b = 80$$

$$b = 20$$

Now, put the value of b in equation (III), we get,

$$a + b = 80$$

$$a + 20 = 80$$

$$a = 60$$

Therefore,

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The number males who qualified = $a = 60$

The number of females who qualified = $b = 20$

The number of males who did not qualify = $7a = (7 \times 60) = 420$

The number of females who did not qualify = $11b = (11 \times 20) = 220$

2019

The total number of candidates who appeared = $(720 - 220) = 500$

The total number of candidates who qualified = 10% of 500 = 50

The total number of candidates who did not qualify = 90% of 500 = 450

Let the number of males who qualified and females who did not qualify be $3c$ and $20c$ respectively.

Also, the ratio of the number of females who qualified and males who did not qualify = $8:100 = 2:25$. Let it be $2d$ and $25d$ respectively.

Now,

$$3c + 2d = 50 \dots\dots\dots(V)$$

And,

$$20c + 25d = 450$$

$$4c + 5d = 90 \dots\dots\dots(VI)$$

By solving equation (V) and (VI), we get,

$$12c + 8d = 200$$

$$12c + 15d = 270$$

$$7d = 70$$

$$d = 10$$

Now, put the value of d in equation (V), we get,

$$3c + 2d = 50$$

$$3c + (2 \times 10) = 50$$

$$3c = 50 - 20$$

$$3c = 30$$

$$c = 10$$

Therefore,

The number males who qualified = $3c = (3 \times 10) = 30$

The number of females who qualified = $2d = (2 \times 10) = 20$

The number of males who did not qualify = $25d = (25 \times 10) = 250$

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The number of females who did not qualify = $20c = (20 \times 10) = 200$

Year	The total number of candidates who appeared	The total number of candidates who qualified	The total number of candidates who did not qualify	The number of males who qualify	The number of females who qualified	The number of males who did not qualify	The number of females who did not qualify
2017	800	200	600	150	50	400	200
2018	720	80	640	60	20	420	220
2019	500	50	450	30	20	250	200
Total	2020	330	1690	240	90	1070	620

Answer: B

Now, according to the question,

2017

Now, the total number of candidates (males + females) who appeared = $(550 + 250) = 800$

$80\% = 800$

$1\% = 10$

$100\% = (10 \times 100) = 1000$

Therefore, the total number of candidates (males + females) who applied = 1000

So, the total number of candidates (males + females) who did not appear = 20% of $1000 = 200$

Now,

The number of males who did not appear = $11X$

So, the number of females who did not appear = $(11X - 20)$

Now,

$11X + (11X - 20) = 200$

$22X = 220$

$X = 10$

Therefore, the value of $\sqrt{(X+6)} = \sqrt{(10+6)} = \sqrt{16} = 4$.

Hence, the correct answer is option B.

12. Questions

2017

The total number of candidates who appeared = 800

The total number of candidates who qualified = $\frac{1}{4}$ of $800 = 200$

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The total number of candidates who did not qualify = $\frac{3}{4}$ of 800 = 600

Now, let the number of males who qualified and females who did not qualify be $3m$ and $4m$ respectively.

Also, the ratio of the number of females who qualified and males who did not qualify = $1:8$. Let it be n and $8n$ respectively. (**Note: $12.50\% = 1/8$**)

So,

$$3m + n = 200 \dots\dots\dots(I)$$

And,

$$4m + 8n = 600$$

$$m + 2n = 150 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$3m + n = 200$$

$$3m + 6n = 450$$

$$5n = 250$$

$$n = 50$$

Now, put the value of n in equation (I), we get,

$$3m + n = 200$$

$$3m + 50 = 200$$

$$3m = 150$$

$$m = 50$$

Therefore,

$$\text{The number males who qualified} = 3m = (3 \times 50) = 150$$

$$\text{The number of females who qualified} = n = 50$$

$$\text{The number of males who did not qualify} = 8n = (8 \times 50) = 400$$

$$\text{The number of females who did not qualify} = 4m = (4 \times 50) = 200$$

2018

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$$\text{The total number of candidates who did not qualify} = (600+40) = 640$$

The ratio of the total number of candidates who qualified and those who did not qualify = $1:8$ (**Note: $11.11\% = 1/9$**)

$$\text{The total number of candidates who qualified} = \{(640/8) \times 1\} = 80$$

$$\text{The total number of candidates who appeared} = (80+640) = 720$$

The ratio of the number of males who qualified and males who did not qualify = $1:7$. Let it be a and $7a$ respectively.

Also, the ratio of the number of females who qualified and females who did not qualify = 1:11. Let it be b and $11b$ respectively.

Now,

$$a + b = 80 \dots\dots\dots(III)$$

$$7a + 11b = 640 \dots\dots\dots(IV)$$

By solving equation (I) and (IV), we get,

$$7a + 7b = 560$$

$$7a + 11b = 640$$

$$4b = 80$$

$$b = 20$$

Now, put the value of b in equation (III), we get,

$$a + b = 80$$

$$a + 20 = 80$$

$$a = 60$$

Therefore,

The number males who qualified = $a = 60$

The number of females who qualified = $b = 20$

The number of males who did not qualify = $7a = (7 \times 60) = 420$

The number of females who did not qualify = $11b = (11 \times 20) = 220$

2019

The total number of candidates who appeared = $(720 - 220) = 500$

The total number of candidates who qualified = 10% of 500 = 50

The total number of candidates who did not qualify = 90% of 500 = 450

Let the number of males who qualified and females who did not qualify be $3c$ and $20c$ respectively.

Also, the ratio of the number of females who qualified and males who did not qualify = 8:100 = 2:25. Let it be $2d$ and $25d$ respectively.

Now,

$$3c + 2d = 50 \dots\dots\dots(V)$$

And,

$$20c + 25d = 450$$

$$4c + 5d = 90 \dots\dots\dots(VI)$$

By solving equation (V) and (VI), we get,

$$12c + 8d = 200$$

$$12c + 15d = 270$$

$$7d = 70$$

$$d = 10$$

Now, put the value of d in equation (V), we get,

$$3c + 2d = 50$$

$$3c + (2 \times 10) = 50$$

$$3c = 50 - 20$$

$$3c = 30$$

$$c = 10$$

Therefore,

$$\text{The number males who qualified} = 3c = (3 \times 10) = 30$$

$$\text{The number of females who qualified} = 2d = (2 \times 10) = 20$$

$$\text{The number of males who did not qualify} = 25d = (25 \times 10) = 250$$

$$\text{The number of females who did not qualify} = 20c = (20 \times 10) = 200$$

Year	The total number of candidates who appeared	The total number of candidates who qualified	The total number of candidates who did not qualify	The number of males who qualify	The number of females who qualified	The number of males who did not qualify	The number of females who did not qualify
2017	800	200	600	150	50	400	200
2018	720	80	640	60	20	420	220
2019	500	50	450	30	20	250	200
Total	2020	330	1690	240	90	1070	620

Answer: A

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2018

$$\text{The number of males who qualified} = 60$$

$$\text{The number of females who qualified} = 20$$

Now, let the number of qualified males and females who are graduates be 15p and 4p respectively.

$$\text{So, the number of qualified males who are post graduates} = (60 - 15p)$$

$$\text{And, the number of qualified females who are post graduates} = (20 - 4p)$$

Now,

$$(60-15p) - (20-4p) = 7$$

$$53 - 15p = 20 - 4p$$

$$11p = 33$$

$$p = 3$$

So, the number of qualified males who are graduates = $15p = (15 \times 3) = 45$

And, the number of qualified females who are graduates = $4p = (4 \times 3) = 12$

Therefore, the total number of qualified candidates (males + females) who are graduates in 2018 = $(45+12) = 57$.

Hence, the correct answer is option A.

13. Questions

2017

The total number of candidates who appeared = 800

The total number of candidates who qualified = $1/4$ of 800 = 200

The total number of candidates who did not qualify = $3/4$ of 800 = 600

Now, let the number of males who qualified and females who did not qualify be $3m$ and $4m$ respectively.

Also, the ratio of the number of females who qualified and males who did not qualify = 1:8. Let it be n and $8n$ respectively. (*Note: 12.50% = 1/8*)

So,

$$3m + n = 200 \dots\dots\dots(I)$$

And,

$$4m + 8n = 600$$

$$m + 2n = 150 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$3m + n = 200$$

$$3m + 6n = 450$$

$$5n = 250$$

$$n = 50$$

Now, put the value of n in equation (I), we get,

$$3m + n = 200$$

$$3m + 50 = 200$$

$$3m = 150$$

$$m = 50$$

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Therefore,

The number males who qualified = $3m = (3 \times 50) = 150$

The number of females who qualified = $n = 50$

The number of males who did not qualify = $8n = (8 \times 50) = 400$

The number of females who did not qualify = $4m = (4 \times 50) = 200$

2018

The total number of candidates who did not qualify = $(600+40) = 640$

The ratio of the total number of candidates who qualified and those who did not qualify = 1:8 (*Note: 11.11% = 1/9*)

The total number of candidates who qualified = $\{(640/8) \times 1\} = 80$

The total number of candidates who appeared = $(80+640) = 720$

The ratio of the number of males who qualified and males who did not qualify = 1:7. Let it be a and 7a respectively.

Also, the ratio of the number of females who qualified and females who did not qualify = 1:11. Let it be b and 11b respectively.

Now,

$$a + b = 80 \dots \dots \dots \text{(III)}$$

$$7a + 11b = 640 \dots \dots \dots \text{(IV)}$$

By solving equation (I) and (IV), we get,

$$7a + 7b = 560$$

$$7a + 11b = 640$$

$$4b = 80$$

$$b = 20$$

Now, put the value of b in equation (III), we get,

$$a + b = 80$$

$$a + 20 = 80$$

$$a = 60$$

Therefore,

The number males who qualified = $a = 60$

The number of females who qualified = $b = 20$

The number of males who did not qualify = $7a = (7 \times 60) = 420$

The number of females who did not qualify = $11b = (11 \times 20) = 220$

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2019

The total number of candidates who appeared = $(720-220) = 500$

The total number of candidates who qualified = 10% of 500 = 50

The total number of candidates who did not qualify = 90% of 500 = 450

Let the number of males who qualified and females who did not qualify be $3c$ and $20c$ respectively.

Also, the ratio of the number of females who qualified and males who did not qualify = $8:100 = 2:25$. Let it be $2d$ and $25d$ respectively.

Now,

$$3c + 2d = 50 \dots\dots\dots(V)$$

And,

$$20c + 25d = 450$$

$$4c + 5d = 90 \dots\dots\dots(VI)$$

By solving equation (V) and (VI), we get,

$$12c + 8d = 200$$

$$12c + 15d = 270$$

$$7d = 70$$

$$d = 10$$

Now, put the value of d in equation (V), we get,

$$3c + 2d = 50$$

$$3c + (2 \times 10) = 50$$

$$3c = 50 - 20$$

$$3c = 30$$

$$c = 10$$

Therefore,

$$\text{The number males who qualified} = 3c = (3 \times 10) = 30$$

$$\text{The number of females who qualified} = 2d = (2 \times 10) = 20$$

$$\text{The number of males who did not qualify} = 25d = (25 \times 10) = 250$$

$$\text{The number of females who did not qualify} = 20c = (20 \times 10) = 200$$

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Year	The total number of candidates who appeared	The total number of candidates who qualified	The total number of candidates who did not qualify	The number of males who qualify	The number of females who qualified	The number of males who did not qualify	The number of females who did not qualify
2017	800	200	600	150	50	400	200
2018	720	80	640	60	20	420	220
2019	500	50	450	30	20	250	200
Total	2020	330	1690	240	90	1070	620

Answer: D

2020

Using statements I and II together

Let the number of males who qualified be t .

So, the number of males who did not qualify = $(t-104)$

The number of females who did not qualify = 24

The number of females who qualified = $(24*5) = 120$

So, the total number of candidates (males + females) who qualified = $(t+120)$

And, the total number of candidates (males + females) who did not qualify = $\{(t-104)+24\} = (t-80)$

Now,

$$(t+120)/(t-80) = 3/1$$

$$t + 120 = 3t - 240$$

$$2t = 360$$

$$t = 180$$

So, the total number of candidates who qualified in the interview in 2020 = $\{(t+120)+(t-80)\} = (2t+40) = (2*180+40) = (360+40) = 400$.

Therefore, both statements I and II together are necessary.

Hence, the correct answer is option D.

14. Questions

2017

The total number of candidates who appeared = 800

The total number of candidates who qualified = $1/4$ of 800 = 200

The total number of candidates who did not qualify = $3/4$ of 800 = 600

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Now, let the number of males who qualified and females who did not qualify be $3m$ and $4m$ respectively.

Also, the ratio of the number of females who qualified and males who did not qualify = $1:8$. Let it be n and $8n$ respectively. (*Note: $12.50\% = 1/8$*)

So,

$$3m + n = 200 \dots\dots\dots (I)$$

And,

$$4m + 8n = 600$$

$$m + 2n = 150 \dots\dots\dots (II)$$

By solving equations (I) and (II), we get,

$$3m + n = 200$$

$$3m + 6n = 450$$

$$5n = 250$$

$$n = 50$$

Now, put the value of n in equation (I), we get,

$$3m + n = 200$$

$$3m + 50 = 200$$

$$3m = 150$$

$$m = 50$$

Therefore,

$$\text{The number males who qualified} = 3m = (3 \times 50) = 150$$

$$\text{The number of females who qualified} = n = 50$$

$$\text{The number of males who did not qualify} = 8n = (8 \times 50) = 400$$

$$\text{The number of females who did not qualify} = 4m = (4 \times 50) = 200$$

2018

$$\text{The total number of candidates who did not qualify} = (600 + 40) = 640$$

The ratio of the total number of candidates who qualified and those who did not qualify = $1:8$ (*Note: $11.11\% = 1/9$*)

$$\text{The total number of candidates who qualified} = \{(640/8) \times 1\} = 80$$

$$\text{The total number of candidates who appeared} = (80 + 640) = 720$$

The ratio of the number of males who qualified and males who did not qualify = $1:7$. Let it be a and $7a$ respectively.

Also, the ratio of the number of females who qualified and females who did not qualify = $1:11$. Let it be b

and 11b respectively.

Now,

$$a + b = 80 \dots\dots\dots(\text{III})$$

$$7a + 11b = 640 \dots\dots\dots(\text{IV})$$

By solving equation (I) and (IV), we get,

$$7a + 7b = 560$$

$$7a + 11b = 640$$

$$4b = 80$$

$$b = 20$$

Now, put the value of b in equation (III), we get,

$$a + b = 80$$

$$a + 20 = 80$$

$$a = 60$$

Therefore,

The number males who qualified = $a = 60$

The number of females who qualified = $b = 20$

The number of males who did not qualify = $7a = (7 \times 60) = 420$

The number of females who did not qualify = $11b = (11 \times 20) = 220$

2019

The total number of candidates who appeared = $(720 - 220) = 500$

The total number of candidates who qualified = 10% of 500 = 50

The total number of candidates who did not qualify = 90% of 500 = 450

Let the number of males who qualified and females who did not qualify be 3c and 20c respectively.

Also, the ratio of the number of females who qualified and males who did not qualify = $8:100 = 2:25$. Let it be 2d and 25d respectively.

Now,

$$3c + 2d = 50 \dots\dots\dots(\text{V})$$

And,

$$20c + 25d = 450$$

$$4c + 5d = 90 \dots\dots\dots(\text{VI})$$

By solving equation (V) and (VI), we get,

$$12c + 8d = 200$$

$$12c + 15d = 270$$

$$7d = 70$$

$$d = 10$$

Now, put the value of d in equation (V), we get,

$$3c + 2d = 50$$

$$3c + (2 \times 10) = 50$$

$$3c = 50 - 20$$

$$3c = 30$$

$$c = 10$$

Therefore,

$$\text{The number males who qualified} = 3c = (3 \times 10) = 30$$

$$\text{The number of females who qualified} = 2d = (2 \times 10) = 20$$

$$\text{The number of males who did not qualify} = 25d = (25 \times 10) = 250$$

$$\text{The number of females who did not qualify} = 20c = (20 \times 10) = 200$$

Year	The total number of candidates who appeared	The total number of candidates who qualified	The total number of candidates who did not qualify	The number of males who qualify	The number of females who qualified	The number of males who did not qualify	The number of females who did not qualify
2017	800	200	600	150	50	400	200
2018	720	80	640	60	20	420	220
2019	500	50	450	30	20	250	200
Total	2020	330	1690	240	90	1070	620

Answer: C

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2019

$$\text{The total number of males who qualified} = 30$$

$$\text{The total number of females who qualified} = 20$$

Now, the ratio of the number of qualified males and qualified females who did not join = 3:5. Let it be 3s and 5s respectively.

$$\text{So, the number of qualified males who joined} = (30 - 3s)$$

$$\text{And, the number of qualified females who joined} = (20 - 5s)$$

Now,

$$(30-3s) - (20-5s) = 12$$

$$18 - 3s = 20 - 5s$$

$$2s = 2$$

$$s = 1$$

So, the number of qualified males who joined = $(30-3s) = (30-3*1) = (30-3) = 27$

And, the number of qualified males who did not joined = $3s = 3$

Therefore, the required ratio = $27:3 = 9:1$.

Hence, the correct answer is option C.

15. Questions

2017

The total number of candidates who appeared = 800

The total number of candidates who qualified = $1/4$ of 800 = 200

The total number of candidates who did not qualify = $3/4$ of 800 = 600

Now, let the number of males who qualified and females who did not qualify be $3m$ and $4m$ respectively.

Also, the ratio of the number of females who qualified and males who did not qualify = $1:8$. Let it be n and $8n$ respectively. (*Note: $12.50\% = 1/8$*)

So,

$$3m + n = 200 \dots\dots\dots(I)$$

And,

$$4m + 8n = 600$$

$$m + 2n = 150 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$3m + n = 200$$

$$3m + 6n = 450$$

$$5n = 250$$

$$n = 50$$

Now, put the value of n in equation (I), we get,

$$3m + n = 200$$

$$3m + 50 = 200$$

$$3m = 150$$

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$$m = 50$$

Therefore,

$$\text{The number males who qualified} = 3m = (3 \times 50) = 150$$

$$\text{The number of females who qualified} = n = 50$$

$$\text{The number of males who did not qualify} = 8n = (8 \times 50) = 400$$

$$\text{The number of females who did not qualify} = 4m = (4 \times 50) = 200$$

2018

$$\text{The total number of candidates who did not qualify} = (600 + 40) = 640$$

The ratio of the total number of candidates who qualified and those who did not qualify = 1:8 (*Note: 11.11% = 1/9*)

$$\text{The total number of candidates who qualified} = \{(640/8) \times 1\} = 80$$

$$\text{The total number of candidates who appeared} = (80 + 640) = 720$$

The ratio of the number of males who qualified and males who did not qualify = 1:7. Let it be a and $7a$ respectively.

Also, the ratio of the number of females who qualified and females who did not qualify = 1:11. Let it be b and $11b$ respectively.

Now,

$$a + b = 80 \dots \dots \dots \text{(III)}$$

$$7a + 11b = 640 \dots \dots \dots \text{(IV)}$$

By solving equation (I) and (IV), we get,

$$7a + 7b = 560$$

$$7a + 11b = 640$$

$$4b = 80$$

$$b = 20$$

Now, put the value of b in equation (III), we get,

$$a + b = 80$$

$$a + 20 = 80$$

$$a = 60$$

Therefore,

$$\text{The number males who qualified} = a = 60$$

$$\text{The number of females who qualified} = b = 20$$

$$\text{The number of males who did not qualify} = 7a = (7 \times 60) = 420$$

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The number of females who did not qualify = $11b = (11 \times 20) = 220$

2019

The total number of candidates who appeared = $(720 - 220) = 500$

The total number of candidates who qualified = 10% of 500 = 50

The total number of candidates who did not qualify = 90% of 500 = 450

Let the number of males who qualified and females who did not qualify be $3c$ and $20c$ respectively.

Also, the ratio of the number of females who qualified and males who did not qualify = $8:100 = 2:25$. Let it be $2d$ and $25d$ respectively.

Now,

$$3c + 2d = 50 \dots\dots\dots(V)$$

And,

$$20c + 25d = 450$$

$$4c + 5d = 90 \dots\dots\dots(VI)$$

By solving equation (V) and (VI), we get,

$$12c + 8d = 200$$

$$12c + 15d = 270$$

$$7d = 70$$

$$d = 10$$

Now, put the value of d in equation (V), we get,

$$3c + 2d = 50$$

$$3c + (2 \times 10) = 50$$

$$3c = 50 - 20$$

$$3c = 30$$

$$c = 10$$

Therefore,

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The number males who qualified = $3c = (3 \times 10) = 30$

The number of females who qualified = $2d = (2 \times 10) = 20$

The number of males who did not qualify = $25d = (25 \times 10) = 250$

The number of females who did not qualify = $20c = (20 \times 10) = 200$

Year	The total number of candidates who appeared	The total number of candidates who qualified	The total number of candidates who did not qualify	The number of males who qualify	The number of females who qualified	The number of males who did not qualify	The number of females who did not qualified
2017	800	200	600	150	50	400	200
2018	720	80	640	60	20	420	220
2019	500	50	450	30	20	250	200
Total	2020	330	1690	240	90	1070	620

Answer: D

2018

The total number of candidates who applied = $(20Y^2+20)$

The number of candidates who appeared = 720

The number of candidates who did not appear = $40Y$

Now,

$$720 + 40Y = 20Y^2 + 20$$

$$20Y^2 - 40Y - 700 = 0$$

$$Y^2 - 2Y - 35 = 0$$

$$Y^2 - 7Y + 5Y - 35 = 0$$

$$Y(Y-7) + 5(Y-7) = 0$$

$$(Y+5)(Y-7) = 0$$

$$Y = -5 \text{ or } +7$$

So, $Y = 7$ (positive integer)

7 is both prime and odd number.

Therefore, question mark = prime and odd both.

Hence, the correct answer is option D.

16. Questions

Answer: A

Quantity I

Article A

The M.P. = Rs. 28000

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The S.P. = Rs. (28000-80X)

The ratio of the C.P and the S.P. = 100:125 = 4:5

The C.P. = $\left[\frac{(28000-80X)}{5} \times 4\right]$ = Rs. (22400-64X)

Article B

The C.P. = Rs. 120X

The M.P. = 150% of 120X = Rs. 180X

The S.P. = Rs. (180X-2000)

Now,

$$(180X-2000) = (22400-64X)$$

$$224X = 24400$$

$$X = 100$$

So, Quantity I = $\sqrt{X} = \sqrt{100} = 10$.

Quantity II

The total number of articles purchased = 4N

The total number of articles sold = $\frac{3}{4}$ of 4N = 3N

The C.P. of each article = Rs. 50

The S.P. of each article = 160% of 50 = Rs. 80

The overall C.P. of all purchased articles = (4N*50) = Rs. 200N

The overall S.P. of all sold articles = (3N*80) = Rs. 240N

Now,

$$240N - 200N = 2000$$

$$40N = 2000$$

$$N = 50$$

So, Quantity II = (N/10) = (50/10) = 5.

Therefore, Quantity I > Quantity II.

Hence, the correct answer is option A.

17. Questions

Answer: B

Quantity I

The total investment made by,

$$A = [(500 \times M) + \{(120\% \text{ of } 500) \times (12-M)\}] = [500M + \{600 \times (12-M)\}] = (500M + 7200 - 600M) = \text{Rs.}$$

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(7200-100M)

$$B = [(300 \times M) + \{(300 \times 2) \times (12 - M)\}] = [300M + \{600 \times (12 - M)\}] = (300M + 7200 - 600M) = \text{Rs. } (7200 - 300M)$$

Now,

$$(7200 - 100M) / (7200 - 300M) = 4/3$$

$$21600 - 300M = 28800 - 1200M$$

$$900M = 7200$$

$$M = 8$$

$$\text{So, Quantity I} = 20M = (20 \times 8) = 160$$

Quantity II

The total investment made by,

$$P = [\{(2N \times 9) + \{(50\% \text{ of } 2N) \times 3\}\}] = \{18N + (N \times 3)\} = (18N + 3N) = 21N$$

$$Q = [\{(2N - 200) \times 10\} + \{(2N - 200 - 300) \times 2\}] = [(20N - 2000) + \{(2N - 500) \times 2\}] = (20N - 2000 + 4N - 1000) = \text{Rs. } (24N - 3000)$$

Now,

$$21N / (24N - 3000) = 7/6$$

$$126N = 168N - 21000$$

$$42N = 21000$$

$$N = 500$$

$$\text{So, Quantity II} = (N - 300) = (500 - 300) = 200$$

Therefore, Quantity I < Quantity II.

Hence, the correct answer is option B.

18. Questions

Answer: B

Quantity I

Let the present ages of husband and wife be 16X years and 15X years respectively.

So, the age of the husband at the time of the marriage = (16X - 5Y)

And, the age of wife at the time of the marriage = (15X - 5Y)

Also, the present age of their son = Y years

Now,

$$(16X - 5Y) + (15X - 5Y) = 52$$

$$31X - 10Y = 52 \dots\dots\dots(i)$$

And,

$$15X - Y = 29 \dots\dots\dots(ii)$$

By solving equations (i) and (ii), we get,

$$465X - 150Y = 780$$

$$465X - 31Y = 899$$

$$119Y = 119$$

$$Y = 1$$

Now, put the value of Y in equation (i), we get,

$$31X - 10Y = 52$$

$$31X - (10 \times 1) = 52$$

$$31X = 52 + 10$$

$$31X = 62$$

$$X = 2$$

So, the age of the husband after 4 years from now = $(16X + 4) = (16 \times 2 + 4) = (32 + 4) = 36$ years.

Therefore, Quantity I = 36

Quantity II

The present age of A = $7Z$ years

The present age of B = $(7Z - 5)$ years

The present age of C = $8Z$ years

Now,

$$7Z + (7Z - 5) + 8Z = (35 \times 3)$$

$$22Z = 105 + 5$$

$$22Z = 110$$

$$Z = 5$$

So, the present age of C = $8Z = (8 \times 5) = 40$ years.

Therefore, Quantity II = 40

So, Quantity I < Quantity II.

Hence, the correct answer is option B.

19. Questions

Answer: E

Quantity I

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The length of train P = 320 meters

The length of train Q = $\frac{3}{4}$ of 320 = 240 meters

The speed of train P = 44 m/sec

The speed of train Q = $\{(44/11)*9\} = 36$ m/sec (*Note: 22.22% = 2/9*)

Now, let the length of the platform be L meters.

So,

$$(320+L)/44 = (240+L)/36$$

$$11520 + 36L = 10560 + 44L$$

$$8L = 960$$

$$L = 120$$

So, the length of the platform = L = 120 meters

Therefore, Quantity I = 120

Quantity II

Let the length of the train be T meters.

Also, let the speed of the train be V m/sec.

Now, when the train crosses the platform,

$$(T+150)/V = 5$$

$$T + 150 = 5V$$

$$5V - T = 150 \dots \dots \dots (i)$$

Also, when the man crosses the train in same direction,

$$T/(V-10) = 10/3$$

$$3T = 10V - 100$$

$$10V - 3T = 100 \dots \dots \dots (ii)$$

By solving equations (i) and (ii), we get,

$$10V - 2T = 300$$

$$10V - 3T = 100$$

$$T = 200$$

Now, put the value of T in equation (i), we get,

$$5V - T = 150$$

$$5V - 200 = 150$$

$$5V = 350$$

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$$V = 70$$

So, 60% of the length of train = 60% of T = 60% of 200 = 120 meters

Therefore, Quantity I = Quantity II.

Hence, the correct answer is option E.

20. Questions

Answer: A

Quantity I

The initial mixture (milk + water) = 8P ml

The initial quantity of milk = $\frac{5}{8}$ of 8P = 5P ml

The initial quantity of water = $\frac{3}{8}$ of 8P = 3P ml

Resultant Mixture

The quantity of milk = 50% of 5P = 2.50P ml

The quantity of water = $\{(50\% \text{ of } 3P) + 20\} = (1.50P + 20)$ ml

Now,

$$2.50P = 1.50P + 20$$

$$P = 20$$

So, Quantity I = 20

Quantity II

The pure quantity of milk initially = 12Q ml

The quantity milk in the resultant mixture = $(12Q - 3Q) = 9Q$ ml

The quantity of water in the resultant mixture = 3Q ml

Now,

$$9Q - 3Q = 60$$

$$6Q = 60$$

$$Q = 10$$

So, Quantity I = 10

Therefore, Quantity I > Quantity II.

Hence, the correct answer is option A.

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